When Earthquakes Cross the Gap: Physics-based Dynamic Modeling of Step-Over Jumps in Normal Faults

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Key points:

- Understanding rupture-jump conditions across normal-fault step-overs remains a challenge for seismic hazard.

- Relative normal-fault segment position governs a trade-off between rupture-jump potential and final magnitude.

- Positive static Coulomb stress changes may overestimate multi-fault rupture by neglecting dynamic effects.

Abstract

Understanding the mechanisms that enable earthquakes to jump across step-overs in fault systems is essential for improving seismic hazard assessment. While many previous studies focused on strike-slip fault systems, the physics of rupture jumping across normal faults remains less investigated. In this study, we perform a series of 3D dynamic rupture simulations to explore the key factors that govern rupture jumps across normal fault segments. By systematically varying critical parameters such as pre-stress levels, principal stress orientation, the distance between emitting and potentially triggered faults, and their relative positions, we identify conditions and dynamic effects that allow ruptures to successfully jump step-overs of up to 5 km. Our findings provide insights into why some dynamically triggered events propagate as self-sustained run-away ruptures, while others are rapidly self-arrested after nucleation. Our results highlight the trade-offs between stress-shadowing effect, relative fault positioning, fault proximity and stopping phases. For instance, fault geometry introduces a fundamental trade-off: closely spaced fault sections enhance rupture-jump potential, yet do not necessarily favor larger magnitudes. For very closely spaced faults, stress-shadowing suppresses slip and limits magnitude, whereas at larger gaps, where the shadow weakens, the rupture can grow and release more energy. Our simulations show that the largest magnitudes occur in configurations where the receiver fault lies on the hanging-wall of the emitter fault at specific overlap and gap distances, highlighting the strong geometric control on multi-fault earthquake behavior for normal faults. Furthermore, the simulations reveal that the static Coulomb stress change approach, commonly used to predict rupture connectivity, does not fully capture the complexities of dynamic rupture processes across normal-fault systems: Dynamically jumping between normal fault segments is more rare than static criteria would suggest.

INTRODUCTION

Historically, geological and seismological observations suggested that earthquake ruptures may not dynamically (during a single event) jump across fault step-overs wider than a few kilometers (typically less than ∼4 km) (Wesnousky, 2006). However, recent complex earthquakes challenge this view, with evidence showing rupture jumps across significantly larger distances. For instance, the 2010 M_w7.2 El Mayor-Cucapah earthquake jumped a step-over of more than 10 km (Wei et al., 2011; Oskin et al., 2012), resulting in a 120 km-long multisegment rupture. Similarly, the 2012 M_w8.6 Sumatra earthquake exhibited a step-over jump of up to 20 km (Meng et al., 2012), or the 2016 M_w7.8 Kaikoura earthquake in New Zealand with rupture jumps of up to 15 km (Hamling et al., 2017).

Previous studies on this topic have focused mainly on strike-slip fault systems, examining the conditions that enable rupture jumps. For example, while the M_w7.1 Düzce earthquake ruptured across multiple fault segments constrained by step-overs less than 2.5 km wide, it terminated at a larger 4–5 km step-over (Duman et al., 2005). Additional investigations have aimed to clarify the physical parameters that enabled the M_w7.8 Kaikoura earthquake to jump across a complex network of strike-slip and thrust faults (Hollingsworth et al., 2017; Ulrich et al., 2019a; Ando and Kaneko, 2018). In particular, the ability of rupture to propagate across fault step-overs of up to ∼15 km may partly reflect structural connections between faults at depth, as suggested by geological and geodetic observations (Nicol et al., 2023).

Dynamic rupture simulators are powerful tools for assessing the viability of specific rupture scenarios. These simulators focused on the coseismic phase, such as FaultMod (Barall, 2009, developed for spontaneous rupture on 3D faults), DGCrack (Tago et al., 2012, based on the discontinuous Galerkin method), and SeisSol (Uphoff et al., 2017; Gabriel et al., 2025, a high-order ADER-DG method for complex rupture dynamics), among others, have been developed to investigate the critical role of parameters such as fault geometry, nucle-

ation depth, and stress orientation in promoting rupture jumps. Several studies employing these dynamic simulators have concentrated on simplified strike-slip fault systems in order to isolate and understand the fundamental governing factors of rupture behavior (e.g., Finzi and Langer, 2012; Bai and Ampuero, 2017). More recently, Hu et al. (2025) showed that the presence of a bi-material interface can substantially modify rupture propagation across step-overs in strike-slip fault systems, increasing the critical jump distance on the compliant side and highlighting the strong influence of material contrast on rupture dynamics and earthquake hazard assessment.

Despite the growing knowledge on strike-slip systems, significantly less attention has been devoted to rupture jumps across normal fault systems. In the context of normal faulting, earlier research has largely focused on structural geology (e.g. fault dip, relay ramps and other linkage processes) and fault growth dynamics. Different approaches have emerged to analyze fault system behavior. Fractal analysis has been employed to study the scaling properties of normal fault populations (Poulimenos, 2000), and stress interaction models have been proposed to assess the effects of fault interactions through perturbations in the surrounding stress field (Gupta and Scholz, 2000). Additionally, kinematic studies such as those by Hikima (2012) and Tanaka et al. (2014) have examined the 11 April 2011 M_w6.6 Hamadori/Iwaki earthquake, which ruptured two subparallel fault segments 2–5 km apart, representing a rare and well-documented case of rupture jumps across step-overs on normal faults. More recently, physics-based modeling efforts have emerged to complement these approaches. For instance, using dynamic rupture modeling Bazán Flores et al. (2024) demonstrated that interactions between the Southern San Andreas Fault and a proximal normal fault beneath the Salton Sea can produce coseismic slip on the normal fault consistent with geological observations, highlighting the potential for dynamic triggering in transtensional step-over zones.

In a complementary line of research, the development of quasi-dynamic simulation tools such as QDYN (Luo et al., 2017), and methods based on elastostatic dislocation theory

(Zielke and Mai, 2023, e.g. MCQsim), has enabled efficient boundary-element simulations of seismic cycles and fault interactions (regardless of the faulting mode). These tools have promoted the exploration of rupture complexity over multiple cycles, including the dynamics of multi-segment ruptures (Galvez et al., 2020) and the emergence of heterogeneous seismicity linked to the presence of damage zones and rheological contrasts within fault structures (Idini and Ampuero, 2022). For example, Rodriguez Piceda et al. (2025) used quasi-dynamic simulations of seismic cycles to examine how fault network geometry and spacing influence recurrence, nucleation, and rupture styles in normal fault systems. Their findings demonstrate that closely spaced across-strike faults tend to promote less periodic seismic cycles and more variable rupture behaviors, while along-strike arrangements have a comparatively milder influence on recurrence patterns. Despite these advances, the application of dynamic rupture simulations to investigate the conditions that enable successful rupture jumps in normal fault systems remains limited, pointing to a need for further development and exploration in this area.

Assessing rupture behavior in normal fault systems is critical, particularly in tectonically active regions such as Nevada (USA) (Wernicke et al., 1988), the Corinth Rift (Greece) (Bell et al., 2009), the East African Rift System (Ebinger and Sleep, 1998), and the Italian Apennines (Ghisetti and Vezzani, 2002). These regions are characterized by damaging seismic activity involving multi-segment normal fault ruptures. For example, the 1980 M_s6.9 Irpinia and the 2016 M_w6.2 Amatrice earthquake ruptured multiple fault segments during single events (Chiaraluce et al., 2017). Understanding how ruptures propagate across step-overs in normal faulting settings is essential for improving seismic hazard assessment.

In this study, we investigate certain physical parameters that may promote rupture jumps in normal fault systems, with particular attention to the distance between fault segments, the pre-stress conditions, and the fault orientation relative to the regional stress field. Using 3D dynamic rupture simulations (SeisSol software, Gabriel et al., 2025), we aim to answer key questions such as: Under what conditions can dynamic rupture successfully

jump from one normal fault segment to an adjacent one? What is the maximum critical distance for successful jumps? How do these conditions affect rupture behavior and resulting earthquake magnitudes? Finally, we explore whether rupture jumps in normal faults can exceed 5 km, as observed in some strike-slip regimes, or whether they are confined to the 4–5 km limit proposed in previous studies (Wesnousky, 2006).

STATIC TRIGGERING AND DYNAMIC RUPTURE-JUMP VIABILITY

A common first-order approach to evaluate whether slip on a given fault segment could be triggered, either instantaneously (i.e., dynamically) or with a delay ranging from days to decades (statically or post-seismically; Freed, 2005), by a preceding rupture on a nearby fault is to perform a Coulomb stress change analysis (King et al., 1994; Stein, 1999). In this framework, the fault hosting the initial rupture is referred to as the emitter fault, whereas the potentially activated neighboring fault is termed the receiver fault. Conceptually, Coulomb stress modeling addresses fault triggering, which is distinct from evaluating the dynamic rupture-jumping potential. The Coulomb approach considers static or quasi-static stress changes after the passage of the rupture front and its associated coseismic wavefield, whereas rupture jumping depends on the transient, high-frequency dynamic stresses required to sustain rupture across fault discontinuities. In this static framework, one estimates the stress changes resolved onto the receiver fault plane caused by slip on the emitter fault. The analysis of these stress changes enables us to determine whether a nearby receiver fault is brought closer to failure. Specifically, by examining the stress transfer from the emitter fault, we can assess if the shear stress (τ) on the receiver fault exceeds its frictional strength, following the Coulomb failure criterion

$$\tau > \mu_s(\sigma_n - P) + C \tag{1}$$

where μ_s is the static coefficient of friction, σ_n is the normal stress, P is the pore fluid pressure, and C is the cohesion of the fault.

However, the static analysis does not account for dynamic effects, such as propagating seismic waves or rapid stress transients, which can co-seismically trigger subsequent ruptures. In a static framework, only two snapshots in time, before and after the potential triggering event, are considered. By examining the stress field and the fault strength at these two instants, and assuming a known geometry and location of a receiver fault, we can assess whether that fault is brought closer to failure. Although Coulomb stress modeling formally addresses fault triggering rather than dynamic rupture jumping, the resulting stress changes on the receiver fault are often used as a first-order proxy for rupture-jump viability (such as in UCERF3, Milner et al., 2022). This approach relies on the assumption that regions with increased stress changes are more likely to be activated during dynamic rupture (i.e., higher jump potential). These areas are identified by quantifying these stress changes through the Coulomb failure function ΔCFF (Rice, 1992), which is defined as

$$\Delta CFF = \Delta\tau - \mu(\Delta\sigma_n - \Delta P) \tag{2}$$

where $\Delta\tau$ is the change in shear stress, $\Delta\sigma_n$ is the change in normal stress and ΔP is the change in pore fluid pressure. Therefore, positive values of ΔCFF indicate that rupture on the emitter fault has brought the receiver fault closer to failure. Classical Coulomb stress studies suggest that far-field static stress changes as small as ~ 0.01 MPa (and up to ~ 0.1 MPa) can bring closer to failure delayed triggering or promote aftershock activity on strike-slip faults (King et al., 1994; Stein, 1999). However, these thresholds correspond to far-field, static triggering and do not directly apply to near-field, dynamic multi-fault rupture cascades, where transient stresses are typically one to two orders of magnitude larger. In this study, the ΔCFF is used only as a first-order proxy for changes in failure potential, not as a predictor of dynamic rupture jumping. In the next section, we introduce

a simplified two-fault model which will serve us to perform our investigation. Results from static analysis and dynamic rupture simulations will be carried out using this model and the different results will help us to explore the rupture jump potential under certain circumstances.

MODEL: GEOMETRY AND PHYSICAL CONDITIONS

Our model setup consisting of two faults is illustrated in Figure 1. For clarity in the following discussion, we define the receiver fault as the one located at the northern boundary of the physical domain, while the triggering fault is always positioned to the south (see Figure 1). The emitter fault is ensured to always dynamically rupture fully, and its rupture may or may not trigger the rupture of the receiver fault. The receiver fault may be dynamically triggered by the rupture of the emitter fault depending on the physical conditions prescribed by a given set of parameters. We will investigate if the initial stress levels, the principal stress orientation and the relative position of the receiver fault with respect to the emitter fault play an important role in dynamically triggering the second rupture. Regarding the fault location, the receiver fault can lie either on the footwall or on the hanging-wall of the emitter fault. This relative positioning can be determined by two distances in the local coordinate system: the overlap (i.e., offset) distance which is measured along the strike direction and the gap distance perpendicular to the strike. A positive gap indicates that the receiver fault lies on the footwall side (FW) of the emitter fault, whereas a negative gap indicates that the receiver fault lies on the hanging-wall side (HW) of the emitter fault. Refer to Figure 1 for details on these distances along the two described directions.

It is necessary to highlight that this geometrical configuration is an oversimplified version of the geometry between Monte Vettore and Laga faults (further details in Section Data and resources). However, this model preserves some of the geometric features and key characteristics of these faults. For instance, the southernmost end point of the receiver fault

coincides with the mapped southernmost end point of Monte Vettore's fault trace (Faure Walker et al., 2021). Another interesting feature kept from the real fault configuration for our dynamic simulations is the elevation model. Thanks to the explored values of gap and offset distances during our simulations, the observed approximately 5 km step-over distance (combining gap and offset distances) between Monte Vettore and Laga faults is also investigated.

[Figure 1 about here.]

Additionally, both faults exhibit the same strike and dip orientations (set to be 339.5^o and 55°, respectively; Figure 1). Each fault in this model has dimensions of 10×10 km, along strike and dip directions, and extends from 0 to 8.19 km depth (according to the dip angle). This fault configuration is selected as a representative setup inspired by well-documented normal fault systems in Central Italy (Faure Walker et al., 2021). Further details on the underlying datasets and fault geometries are provided in the Data and resources section.

Under this assumption, both faults are considered to be subjected to the same regional stress, with a principal orientation aligned with the fault's orientation (SH_{max} =339.5^o azimuth, Figure 1). The faults are embedded in a layered medium based on a 1D velocity structure appropriate for the Norcia-Amatrice region (see Section Data and resources and Table S1 Tinti et al., 2021; Taufiqurrahman et al., 2022). In addition, the digital elevation model for this region (see Table S2) is incorporated to account for realistic topography in our model. Both the velocity and elevation models are considered for estimating the depth-dependent background stress field. A depth-dependent high pre-stress level is imposed, with both faults nearing failure, characterized by $S = 0.1$ (S being the dimensionless parameter representing fault strength, see Figure 2), where

$$S = \frac{\tau_p - \tau_0}{\tau_0 - \tau_r}, \tag{3}$$

being τ_p, τ_r and τ_0 the peak, residual and initial fault strength levels, respectively. It is

assumed that both, peak and residual strength levels (see Figure 2) are given as

$$\begin{aligned} \tau_p &= C + \mu_s \sigma_n, \\ \tau_r &= \mu_d \sigma_n. \end{aligned} \tag{4}$$

In this study, the initial stress level is prescribed using the systematic approach introduced by Ulrich et al. (2019a). This approach incorporates Mohr–Coulomb theory, fault fluid pressurization, and optimizes the alignment between fault shear traction and slip vectors to ensure realistic stress configurations for sustained rupture propagation. We apply this framework to ensure consistency with regional stress orientations and fault slip in our model. The initial stress levels are set by leveraging a stress state constrained by the background regional stress conditions, fluid pressure and a depth-dependent stress function (see Ulrich et al., 2019a, for details on the seven independent parameters). This setup allows us to generate an initial stress field aligned with our fault geometry and realistic dynamic rupture scenarios (see Figure 2).

While the Ulrich et al. (2019b) model incorporates complex overpressurized fault zones (required to study the 2016 Kaikoura earthquake), we assumed a simpler configuration considering only the rheological static load, and depth-dependent linear cohesion and pore pressure functions. The cohesion is prescribed to vary linearly from 2 MPa at the surface (0 km depth) to 0 MPa at 2 km depth, and it remains fixed at 0 MPa at greater depths. This parameterization inhibits rupture initiation at shallow depth. While for the pore pressure, we only assumed a linear depth-dependent water pressure considering a constant water density of 1000 kg/m^3. Then, the vertical stress component is defined as

$$\sigma_{zz} = (\rho - \rho_w) * g * \min(z_{min}, z), \tag{5}$$

where, ρ and z represent the local density and depth at a given location on the fault, while ρ_w and z_{min} correspond to constant values for the water density (1000 kg/m^3) and a

minimum depth (1.5 km) used to define a fixed static load for the uppermost 1.5 km (from 0 to 1.5 km depth, without accounting for topography), and the constant g denotes the gravitational acceleration. Figure 2 illustrates the initial shear stress state along the dip direction (principal direction for normal faults) prescribed based on these parameters for a case where the receiver fault lies on the hanging-wall of the emitter fault and which offset and gap distances are set to -2.5 (25% of overlap between faults along the strike direction) and -5.0 km, respectively.

[Figure 2 about here.]

For all the following simulations, we assume a linear slip-weakening friction law. In this constitutive model, fault strength decreases from the static friction coefficient μ_s to the dynamic friction coefficient μ_d over the prescribed critical slip-weakening distance d_c. In our simulations, we prescribe a fixed and homogeneous critical slip-weakening distance ($d_c = 0.15$ m) over both fault segments. To initiate the first rupture, we employ the forced-rupture-time approach, which provides a more controlled and stable nucleation process than standard spontaneous nucleation with the linear slip-weakening formulation (see SeisSol official documentation for details, Gabriel et al., 2025). The forced rupture is applied only within the nucleation region at the center of the emitter fault, where the friction coefficient is smoothly reduced from $\mu_s = 0.6$ to $\mu_d = 0.4$ over a period of 1 s inside a spherical region of 0.5 km radius (see Section Data and resources for further details). Rupture initiates once the initial shear stress exceeds the evolving Coulomb failure threshold, which is updated at each timestep. This smooth nucleation procedure minimizes spurious oscillations and numerical artifacts associated with abrupt rupture initiation and follows the recommendations of the SCEC/USGS Dynamic Rupture Code Verification exercises TPV29 and TPV30 (Harris et al., 2018), and has been adopted in previous dynamic rupture studies using SeisSol (e.g., Ulrich et al., 2019a).

Because this study is focused on normal faulting, the principal stress component is

vertical ($\sigma_{zz} = s_1$). Based on that, the other principal stress magnitudes (s_2 and s_3) are prescribed by the relative fault strength S (equation 3) according to the relative pre-stress ratio R, where $R = 1/(1+S)$, and the stress-shape ratio $s2ratio = (s_2 - s_3)/(s_1 - s_3)$, where $s_1 > s_2 > s_3$ are the principal stress magnitudes. This stress parameterization provides a consistent initial stress state (see Figure 2) and ensures that the modeled faults begin in a physically plausible pre-failure regime suitable for dynamic rupture simulations.

STATIC COULOMB STRESS CHANGE ANALYSIS

To illustrate the results and potential limitations of a static Coulomb stress change analysis, we investigate two different scenarios using our simplified two-fault model. In both cases, the receiver fault (northern fault) remains fixed in space and is located either on the hanging-wall (Figures 3a,c) or on the footwall (Figures 3b,d) of the emitter fault (southern triggering fault), whose position varies between the two configurations. In both scenarios, the faults are separated by 5 km (gap distance), with a 25% overlap between them (offset distance). To estimate the static stress change induced by the rupture on the emitter fault in both cases, we apply equation 2 with a static friction coefficient $\mu_s = 500$ on the receiver fault. This friction coefficient ensures no interaction with the rupture's evolution, helping us identify potential triggering zones on the hypothetically locked receiver fault.

Static analyses of the stress changes induced on the receiver fault by a simulated earthquake on the emitting fault are performed for both hanging-wall and footwall configurations (Figure 3a-d). Panels (a) and (b) illustrate the two 3D fault configurations in map view, together with the Coulomb stress change evaluated on the emitter and receiver faults (solid colors) and on a horizontal plane at 2 km depth (transparent colors). Panels (c) and (d) show the corresponding Coulomb stress change resolved on the receiver fault plane for the footwall and hanging-wall configurations, respectively. The first result highlighted by the static analysis for these two extreme cases (i.e., $\pm$5 km gap distance) is that the relative

position of the receiver fault plane significantly influences the expected effect of the emitter fault's rupture. The static Coulomb stress-change pattern is markedly asymmetric between the two configurations: when the receiver fault is on the footwall of the emitter fault, positive Coulomb stress changes are concentrated at greater depths (Figure 3c), whereas in the hanging-wall configuration, positive Coulomb stress changes are concentrated at shallower depths (Figure 3d). This contrast reflects the asymmetry of the static Coulomb stress-change field around the dipping emitter fault and its intersection with the receiver fault at different positions within that field.

However, the static stress-change results shown in Figures 3c and 3d indicate that, in both cases, rupture on the emitter fault increases the failure potential on the northern portion of the receiver fault. Particularly, the Coulomb stress changes are strongly positive on the northern side ($\Delta CFF > 0.2$ MPa; see bottom panels in Figure 3). While static Coulomb modeling does not predict whether a dynamic rupture will actually nucleate or continue as a run-away event, these significantly positive stress changes might provide a first-order indication that this northern region on the receiver fault is more likely to be dynamically activated if rupture conditions are favorable. In contrast, both cases exhibit a pronounced zone of negative ΔCFF near the southern 2.5 km overlap between the two faults. This zone corresponds to a static stress-shadow region. Such stress-shadowing is an expected effect of the overlapping fault geometry.

[Figure 3 about here.]

Could the impact of the emitter fault rupture on the receiver fault be appropriately estimated using only static stress change analysis? This is a valid concern, as there are observed cases where normal faults rupture without dynamically activating nearby fault segments, even when they are less than 10 km apart. A well-documented example includes the 2016 Central Italy sequence where three main events (24/08/16 Mw6.2 Amatrice, 26/10/16 Mw6.1 Visso and 30/10/16 Mw6.6 Norcia earthquakes, USGS catalog last

accessed on 01/11/25) ruptured distinct fault segments despite their close spacing (Chiaraluce et al., 2017; Improta et al., 2019a). Could it be that the static analysis overlooks other important factors that control rupture jumps and the ability of subsequent triggered ruptures to sustain their propagation? To address these questions, the next section presents several dynamic rupture simulations under different initial conditions.

DYNAMIC ANALYSIS

Different scenarios are set to investigate the interaction between the two normal faults from the simplified configuration described before. The geometrical, physical conditions of the model and the friction law used to explore the rupture jump potential through dynamic rupture simulations are as described in Section Model: geometry and physical conditions. In this setting, during the first rupture at least three scenarios are possible: (1) no coseismic rupture jump occurs across the step-over because the dynamic stress perturbations are insufficient to overcome the failure threshold on the receiver fault; (2) the rupture successfully jumps and propagates in a run-away manner, fully rupturing the receiver fault; or (3) the rupture successfully jumps across the step-over and initiates slip on the receiver fault, but then self-arrests because the available energy is insufficient to sustain further propagation.

To explore the effects of different relative positions between the two faults, we fixed the position of the receiver fault (i.e. the receiver of the potential rupture jump) while varying the location of the first fault (the emitter) for each simulation scenario. A systematic approach is used to define the relative positioning of the faults based on two parameters: offset and gap distances (Section Model: geometry and physical conditions and Figure 1). Both distances are referenced from the southernmost, shallowest endpoint of the receiver fault in a local coordinate system. A positive offset indicates that the emitter fault is located farther along the strike direction relative to the reference point of the receiver fault (no overlap exists between them), while a negative offset means the two faults overlap along

the strike direction. This setup allows for a systematic analysis of the effects of relative fault positioning on rupture dynamics. In order to explore the impact of these geometrical parameters, in our rupture scenarios the offset and gap distances vary between -5.0 to +5.0 km along both directions (see Table 1 for details). It is worth mentioning that the different values of gap distance explored are relevant to investigate if the relative position of the receiver fault (i.e. being on the footwall or hanging-wall of the emitter one) inhibits or promotes the rupture jump across the step-over.

The influence of the direction of the second principal stress component (SH_{max}) on the rupture jump potential is also investigated. The direction of SH_{max} is varied from 334.5^o to 344.5^o in increments of 5°, with the optimal direction identified as 339.5^o, which aligns with the strike direction of the faults. In addition, different pre-stress levels (hence different stress-drop) are examined by varying the S parameter from 0.1 (highest level) to 0.3 (lowest level) in increments of 0.1. All the parameters and their respective ranges explored in the simulations are summarized in Table 1.

[Table 1 about here.]

A total of 495 distinct scenarios are constructed by combining five offsets, eleven gap distances, three pre-stress levels, and three $SH_{\max}$ orientations (see Table 1). Among these, 27 cases correspond to configurations in which the two fault segments lie on the same physical plane (i.e., gap distances of 0 km and negative offsets). These cases are excluded, leaving 468 scenarios investigated in this study. All simulations are conducted using the simplified configuration described in Section Model: geometry and physical conditions. The rupture on the emitter fault consistently nucleates as described in Section Model: geometry and physical conditions and fully propagates across the fault. For each simulated scenario, we estimate the moment magnitude (M_w). When a successful rupture jump occurs, the total ruptured area along both faults is considered to estimate the final magnitude, which serves as a proxy for the total energy released. The results presented below describe the

different effects identified in this exploration. All simulations are performed using SeisSol (Gabriel et al., 2025). Complete details on the input files and simulation results are provided in Section Data and resources.

RESULTS

Figure 4 summarizes the results of 156 simulations, where $SH_{max} = 339.5°$ is aligned with the optimal orientation, coinciding with the fault's strike direction. The results from the remaining 312 cases (when $SH_{max} = 334.5°$ and $SH_{max} = 344.5°$) are included in the Supplementary Material. A threshold of 1 cm for the final slip on the receiver fault is considered to determine whether the receiver fault slipped or not by the end of the simulated ruptures (after 20 s). If this threshold is exceeded, the case is classified as a scenario with a successful rupture jump. The choice of a 1 cm threshold is not critical, as the classification of successful versus unsuccessful rupture jumps remains robust for reasonable variations of this value (1 to 5 cm). This threshold was selected as a practical criterion to distinguish negligible slip from meaningful rupture activation on the receiver fault. Regarding the final behavior observed across the entire set of simulations, the scenarios can be subdivided into three main groups. First, there are scenarios where the rupture fully propagates only on the emitter fault plane, without producing a step-over jump (empty circles in Figure 4). Second, some scenarios result in complete rupture of both fault planes, where successful step-over jumps occur followed by run-away behaviors (filled circles in Figure 4). Finally, a third group includes scenarios where step-over jumps occur succesfully, but the resulting triggered second rupture is self-arrested before propagating across the whole receiving fault plan (half-filled circles in Figure 4).

For the optimal stress orientation ($SH_{max} = 339.5°$), corresponding to all the scenarios shown in Figure 4, 76 out of the 156 simulations (48.7%) exhibited a successful rupture jump. Among these successful jumps, 66 cases (42.3% of the total ensemble) evolved into

sustained run-away ruptures, whereas 10 cases (6.4%) self-arrested shortly after nucleation. In contrast, 80 simulations (51.3%) did not trigger any rupture on the receiver fault. The results for the other regional stress orientations are summarized in the Supplementary Material.

[Figure 4 about here.]

Geometrical effects

The asymmetric behavior (previously mentioned in Section Static Coulomb stress change analysis) is observed, attributed to the different regimes experienced by the receiver fault depending on the relative position between both segments. When the receiver fault is located in the hanging-wall block of the emitter fault, the rupture initiated by the emitter fault is more likely to propagate to the receiver fault. Conversely, when the receiver fault is situated in the footwall block of the emitter fault, the likelihood of a successful rupture jump is significantly reduced, and the occurrence of a triggered second rupture with a run-away behavior becomes even less probable. For instance, in Figure 4 when faults overlap (looking at offsets < 0 km), it is observed that all 30 cases (100%), where the receiver fault lies on the hanging-wall (HW) of the emitter fault, exhibited a successful rupture jump. 6 out of those 30 cases (20%) display a self-arrest rupture behavior, the remaining 24 cases present run-away ruptures. In contrast, when the receiver fault is located on the footwall (FW) of the emitter fault and there is an overlap between faults, only 17 out of 30 cases (56%) experience a successful rupture jump, with 4 (13%) exhibiting a self-arrest rupture behavior. The same asymmetric behavior can be observed when there is no overlap between faults (see Figure 4 at offsets ≥ 0 km). Examples of this asymmetric behavior can be clearly observed in Figures S3 and S4 of the Supplementary Material.

A direct relationship is observed between fault proximity and rupture-jump potential. In Figure 4, it is shown that all 16 scenarios (100%) within an absolute gap distance of 1 km (HW and FW) and with a 0 (no overlap) or negative offset (with overlap) exhibited

successful rupture jumps followed by run-away behavior. This pattern persists even at low pre-stress levels ($S = 0.3$). In contrast, at absolute gap distances of 5 km and 0 or negative offset values, only 12 out of the 16 scenarios (75%) exhibit successful rupture jumps, with 6 (37%) of them resulting in self-arrested ruptures. Remarkably, at low pre-stress levels the rupture jump does not occur when the receiver fault is on the footwall of the emitter fault beyond gap distances of 2 km (for $S = 0.2$) and 1 km (for $S = 0.3$). This proximity-related behavior is strongly emphasized when there is no overlap between faults (at positive offset values in Figure 4). For instance, 4 (11%) out of the 44 cases with 2.5 and 5.0 km offset distances (no overlap) and low pre-stress levels ($S > 0.1$) exhibit successful jumps, but the triggered second ruptures present a self-arrested behavior.

The distribution of the estimated moment magnitudes for the 156 scenarios sharing the optimal regional stress orientation ($SH_{max} = 339.5°$) is represented by the color scale in Figure 4 (see Tables S3-S8 for additional details). The seismic moment is computed by integrating the modeled slip over the portion of the fault where the slip exceeds 1 cm, and the corresponding moment magnitude is then derived from this seismic moment. Notably, the relationship between fault proximity and the resulting magnitudes is not straightforward because of the stress-shadowing effect. A first clear pattern emerges as the distance between faults (controlled by the gap and/or offset parameters) increases: larger distances amplify the magnitude differences between HW and FW configurations. For example, at 0.0 km offset and small gap distances, the magnitudes are nearly identical ($M_w = 6.52$ at a gap -1.0 km and $M_w = 6.51$ at a gap +1.0 km, Table S3). In contrast, for the same 0.0 km offset but a larger separation, the difference becomes more pronounced, with $M_w = 6.55$ at gap values of –5.0 km and $M_w = 6.43$ at +5.0 km, respectively. This growing asymmetry between HW and FW cases appears consistently across all pre-stress levels and configurations (see Figures S3 and S4 for more details), although it is less pronounced when the faults do not overlap (Tables S3 to S8).

In addition, when the two faults overlap, we consistently observe larger magnitudes in

the HW cases (gap < 0 km) compared to the FW cases (Tables S3 and S4). For instance, in the 5.0 km offset configuration (Figure S4), the HW scenario at -5.0 km gap produces magnitudes of $M_w = 6.55$, 6.52, and 6.49 for $S = 0.1$, 0.2, and 0.3, respectively. In contrast, the corresponding FW cases at +5.0 km gap yield lower values of $M_w = 6.43$, 6.41, and 6.39 (Tables S3 and S4). A similar pattern is seen in the +5.0 km offset configuration: at 5.0 km gap, magnitudes reach $M_w = 6.43$, 6.40, and 6.38 for $S = 0.1$, 0.2, and 0.3, while at +5.0 km gap they remain slightly higher but still within the FW range ($M_w = 6.43$, 6.41, and 6.39). Therfore, these results show that, when it is present, the stress-shadowing effect systematically reduces the magnitudes in FW configurations, whereas HW cases receive more favorable loading to produce larger magnitudes.

It is also important to note that, for all pre-stress levels, the scenarios with the largest magnitudes correspond to HW configurations (Tables S3, S4 and S5). At $S = 0.1$, the maximum magnitude is $M_w = 6.56$ (offset –2.5 km, gap –4 km); at $S = 0.2$, it is $M_w = 6.53$ (offset 0 km, gap –4 km); and at $S = 0.3$, it is $M_w = 6.49$ (offset –5 km, gap –5 km). These maximum-magnitude cases do not correspond to the same geometric configuration across the three stress levels, nor are they necessarily the ones with the smallest separation between fault segments. For comparison, the maximum magnitudes among FW scenarios are systematically lower: at $S = 0.1$, the largest FW event, $M_w = 6.55$, occurs at offset 0 km and gap +4 km; at $S = 0.2$, the FW maximum is $M_w = 6.49$ (offset 0 km, gap +1 km); and at $S = 0.3$, it is $M_w = 6.46$ (also offset 0 km, gap +1 km). Interestingly, the HW scenarios that produce the largest magnitudes differ across pre-stress levels, and the same holds for the FW cases (with the exception of cases with $S = 0.2$ and $S = 0.3$).

Dynamic effects

To investigate why, in some cases, a rupture can successfully jump and subsequently exhibit run-away behavior on the receiver fault, whereas in other cases the triggered secondary rupture is self-arresting, two specific scenarios are analyzed in detail. Both scenarios share an offset of -2.5 km (25% fault area overlap) but differ in their gap distances: the

first has a gap of +5.0 km, while the second has a gap of −5.0 km. These two scenarios were selected for their contrasting geometries (i.e., footwall versus hanging-wall), rupture jump potential, and differing outcomes: the first scenario (gap of +5.0 km, FW case) exhibits a successful rupture jump followed by a self-arrested rupture, while the second (gap of -5.0 km, HW case) demonstrates run-away behavior after a successful jump. Figures 5, 6, and 7 (and Videos S1 and S2) illustrate these two scenarios at three representative time steps: 3.0, 4.2, and 6.4 s, respectively. All three figures follow the same layout, with the footwall (panels (a), (c), and (e)) and hanging-wall (panels (b), (d), and (f)) configurations shown in the left and right columns, respectively. The upper panels show map views of the evolving rupture, with fault slip represented on the faults by solid colors and the wavefield, expressed as particle velocity, by transparent grayscale shading, at 2 km depth. The middle panels ((c) and (d)) show vertical cross-sections along the A-B profiles indicated in the map views (black dashed line in (a) and (b)). These cross-sections highlight the interaction of the propagating wavefield with the two fault planes, whose intersections with the cross-section are indicated by blue and red lines for the emitter and receiver faults, respectively. The lower panels ((e) and (f)) show the corresponding Coulomb stress change resolved on the receiver fault, allowing the evolution of dynamic stress transfer to be directly related to the rupture response of the receiver fault.

For the case where the receiver fault lies on the hanging-wall of the emitter fault, the second run-away rupture is ultimately triggered at 4.2 s (see Figure 6). However, a weaker preliminary rupture is initiated earlier, between 2.8 s and 3.0 s (Figure 5), but it remains small and stops shortly after its nucleation. This initial rupture is triggered by the arrival of the S-wave front coming from the emitter fault (see Figure 5b,d). As shown in Figure 5b, the shear stress on the receiver fault (hanging-wall of the emitter fault) first increases and then drops in the upper southern corner. This behavior corresponds to the intersection of the S-wave front with the fault lying on the hanging-wall. The presence of the S-wave front is visible both in the horizontal map view (Figure 5b) and in the vertical profile, where a

positive vertical particle velocity field forms a half-moon shape (Figure 5d). The S-wave front is associated with an increase in the dynamic Coulomb stress change at the moment the preliminary rupture is triggered (Figure 5f). However, this initial positive stress change is followed by a negative change over time, which prevents the further development of the rupture (Supplementary Material Video S1).

[Figure 5 about here.]

[Figure 6 about here.]

The self-sustaining triggered rupture on the receiver fault lying on the hanging-wall initiates at 4.2 s (Figure 5b) in an area close to both the emitter fault and the free surface. This location lies within a broad region of negative Coulomb stress change, as shown in Figure 6f. However, at 4.2 s, a localized patch of positive Coulomb stress change emerges. This increase in Coulomb stress is associated with a stopping phase coming from the northern edge of the emitter (southern) fault. The origin is evident, as it forms a circular pattern around the fault edge, clearly visible in the map view of Figure 5b. At this moment, the localized increase in Coulomb stress change is sufficient (qualitatively) to initiate a spontaneous rupture on the receiver fault, near the shallow boundary between the first and second velocity layers (approximately 4.5 km depth, see the Supplementary Material for details).

On the other hand, for the case where the receiver fault lies on the footwall of the emitter fault, the rupture on the receiver fault is triggered at 6.4 s (Figure 7). In this footwall scenario, the receiver fault experiences different wave arrivals compared to the hanging-wall case due to asymmetry. First, the particle velocity motion on the footwall is in the opposite direction (Figures 5a,c). The upward particle-velocity wave hitting the fault is associated with a reduction in shear stress and a negative Coulomb stress change (Figure 5e). Second, the radiation from the emitter (southern) fault reaches the bottom of the receiver fault first (Figures 5a,c)). As a result, the impact of these incoming waves is less pronounced on the shallower part of the receiver fault (the area closest to failure) compared to the hanging-wall

case. This difference is primarily due to the varying geometrical distances and the different phases of the radiated waves.

[Figure 7 about here.]

From the 5.8 s snapshot (Supplementary Material Video S2a,e), in the bottom and top left panels, we can observe that the stopping phase propagating through the footwall block crosses the receiver fault without triggering the rupture. Between 6.0 and 9.0 s, the left panels show that this strong stopping phase is followed by a free surface reflection phase, represented by black colors at shallow depths (see Figure 7a,c and Supplementary Material Video S2a,c). This reflected wave propagates downward and outward from the vertical cross-section plot (Figure 7c). It appears that the reflected phase from the free surface is responsible for triggering the rupture on the receiver fault at approximately 7.0 s. This triggered rupture begins near the northern, shallower edge of the receiver fault. However, due to its insufficient energy budget (qualitatively small) and its nucleation location, this triggered rupture is unable to sustain a larger rupture propagating southward through areas where the stress field has already been disturbed by the first rupture and its dynamic-static effects, ending up with an arrested behavior.

DISCUSSION

Modeling assumptions and implications

An important assumption of the present study is the use of a linear slip-weakening (SW) friction law. Previous comparisons between SW and rate-and-state friction (RSF) formulations have shown that SW tends to favor larger immediate rupture jumps because failure occurs abruptly once the weakening threshold is reached, whereas RSF introduces time- and stress-dependent nucleation processes that can inhibit some immediate jumps (e.g., Kroll et al., 2023). At the same time, RSF permits delayed rupture jumps that are not captured by SW formulations. Consequently, the critical jump distance of 5 km identified here

should be interpreted within the framework of immediate dynamic rupture transfer modeled using the SW law, for the configurations tested here (planar, parallel fault segments and the explored stress conditions), and may represent an upper-bound estimate. Nevertheless, Kroll et al. (2023) showed that the overall rupture patterns, renucleation locations, and first-order geometric controls remain broadly similar between the two friction formulations. We therefore expect the primary conclusions regarding the influence of fault overlap, gap distance, hanging-wall versus footwall configuration, and stress-shadowing effects to remain robust.

Beyond the simplified fault geometries considered in this study, previous dynamic rupture studies suggest that other geometric parameters, such as fault length and dip, may also influence rupture transfer. Larger fault segments can host larger ruptures and generate stronger dynamic stress perturbations, potentially facilitating rupture transfer across larger step-overs. In such cases the distance between the rupture nucleation point and the fault junction should also be explored. Previous studies of strike-slip faults have shown that this distance influences the rupture inertia available when the rupture reaches the discontinuity, thereby affecting the likelihood of a successful rupture jump (Bai and Ampuero, 2017). Likewise, fault dip can modify the dynamic stress field, rupture radiation pattern, and free-surface interactions. Depending on the regional stress orientation and fault geometry, variations in fault dip may either facilitate or inhibit rupture transfer by altering dynamic stress concentrations and rupture propagation across fault discontinuities (e.g., Marschall and Douilly, 2025; Peshette et al., 2019). These effects are therefore likely to be strongly configuration dependent. A systematic investigation of these parameters would considerably increase the parameter space and is beyond the scope of the present study. Natural normal faults exhibit frictional and geometrical complexities that may locally enhance or inhibit rupture transfer across step-overs by modifying dynamic stress concentrations (Shi and Day, 2013; Bruhat et al., 2016; Taufiqurrahman et al., 2022). Such complexities could therefore also modify the stress-shadowing effects identified here and, consequently, their

influence on rupture jump potential and earthquake magnitude. The interaction between fault complexities and stress shadowing would be a topic for future investigations.

Static vs. dynamic rupture transfer

In Section Static Coulomb stress change analysis, the static Coulomb stress-change imposed on a receiver fault by an initial rupture on an emitter fault is calculated as a reference for assessing the potential for dynamic triggering. The analysis shows that, for both the footwall and hanging-wall configurations (offset of -2.5 km and gap of ± 5 km; Figure 3a,b), the receiver fault experiences both negative (stress-shadow) and positive Coulomb stress changes with broad regions of $\Delta CFF > 0.1$ MPa (Figure 3c,d). Nevertheless, the footwall configuration exhibits a larger area of negative Coulomb stress-change than the hanging-wall configuration. Based solely on the static Coulomb stress change, multi-fault ruptures would therefore be expected in both configurations. The dynamic simulations, however, reveal a more complex behavior. For the optimal stress orientation ($SH_{max} = 339.5°$; Figure 4) only 48.7% of the dynamic simulations produced a successful rupture jump, despite the presence of broad regions of positive static ΔCFF on the receiver fault (e.g. Figure 3c,d). Furthermore, only 42.3% of the simulations developed sustained run-away ruptures, while 6.4% resulted in self-arrested triggered ruptures. These results demonstrate that positive static ΔCFF indicates potentially favorable conditions for rupture transfer but does not reliably predict whether a dynamic rupture jump will occur or, if initiated, whether the secondary rupture will self-arrest or evolve into a run-away rupture. This limitation arises because static analysis neglects transient stress perturbations, wave interactions, stopping phases, rupture nucleation conditions, and the evolving energy budget that ultimately governs rupture propagation across fault discontinuities.

HW-FW Asymmetry in rupture transfer

A key result emerging from the dynamic simulations is the pronounced asymmetry in rupture-jump propensity between hanging-wall and footwall configurations. This asymmetry likely reflects the combined effects of static stress transfer, asymmetric wave radiation,

and their influence on rupture nucleation on the receiver fault. First, the static Coulomb stress-change distribution is asymmetric, with a broader region of negative Coulomb stress change on the footwall side of the emitter fault (Figures 3a,b). This asymmetry is also expressed on the receiver fault, where the stress shadow extends to shallower depths in the FW configuration, whereas the strongest negative Coulomb stress changes are concentrated at greater depths in the HW configuration (Figures 3c,d). These differences arise from the geometrical configuration of the faults within the regional stress field and may contribute to the observed asymmetry in rupture-jump propensity. Second, the radiated wavefield is also asymmetric, as shown by the vertical particle-velocity snapshots (Figures 5-7) and supplementary videos. Although individual surface-reflected phases cannot be clearly identified, the wavefront propagating behind the emitter fault is smoother than that emitted in front of it, which exhibits alternating upward and downward particle velocities (Figures 5a,b and 6a,b). This difference may reflect an asymmetric interaction of the radiated wavefield with the free surface. Along the depth profile, the wavefield propagates preferentially toward the surface on the HW side and toward greater depths on the FW side of the emitter fault (Figures 5c,d and Videos S1 and S2). Consequently, the receiver fault is first affected at shallower depths in the HW configuration and at greater depths in the FW configuration. Thus, the dip of the emitter fault within the half-space produces an asymmetric wavefield that interacts with different portions of the receiver fault. Third, this asymmetric dynamic loading is reflected in the rupture behavior (Section Dynamic analysis and Figures 5, 6, and 7). In the HW configuration, rupture triggering on the receiver fault is associated with transient seismic waves and the resulting rapidly evolving Coulomb stress changes. An initial rupture nucleates but rapidly arrests, whereas a second nucleation subsequently develops into a sustained run-away rupture at the boundary between the static stress-shadow zone and the loaded region (Figure 6f). In contrast, in the FW configuration, no run-away rupture is triggered despite significant positive static Coulomb stress change (Figure 3c). Instead, a small rupture initiates farther north on the receiver fault but does not develop

into sustained propagation. One possible explanation is that the shallow portion of the receiver fault is more sensitive to small positive Coulomb stress perturbations because of the lower normal stress at shallow depth. In the FW configuration, however, the first strong wave arrivals predominantly affect deeper portions of the receiver fault, where higher normal stresses may make dynamic triggering more difficult. In addition, the static stress shadow near the fault termination may favor nucleation farther along the receiver fault, outside the negatively loaded region. Fourth, the nucleation locations provide additional insight into the triggering mechanism. Across the simulations, rupture tends to nucleate on the receiver fault near the termination of the emitter fault (Figures 5, 6, and Videos S1 and S2). This spatial association suggests that the stopping phase radiated from the emitter-fault termination may contribute to raising the local stress above the failure threshold. For larger gap distances in overlapping configurations, a different wave phase may contribute to triggering the jump, as observed in the FW case shown in Figure 7. Taken together, these observations suggest that the HW-FW asymmetry in rupture-jump propensity cannot be attributed to static stress transfer alone. Instead, it likely emerges from the combined effects of asymmetric static Coulomb stress changes and dynamic loading by the radiated wavefield, which control where and when favorable nucleation conditions develop on the receiver fault.

Rupture transfer in the Laga-Monte Vettore fault system

Both static and dynamic Coulomb stress analysis consider the stress-shadowing effect. However, the final magnitude of a rupture will depend on whether the dynamic Coulomb stress configuration is favorable for a rupture jump. The partial rupture of the Monte Vettore Fault during the 24/08/16 Mw6.2 Amatrice earthquake is an interesting example to discuss. The Amatrice earthquake ruptured two distinct shallow asperities (depth < 6 km) over a total rupture length of approximately 25 km (Improta et al., 2019): the southern asperity (the Laga "emitter" Fault) and the northern asperity (the southern termination of the Monte Vettore "receiver" fault in the footwall of the Laga Fault) (Faure Walker

et al., 2021). In the coordinate system defined in Section Model: geometry and physical conditions, this configuration corresponds to a positive gap distance of 3 to 5 km and a positive along-strike offset of 2 to 3 km, with no overlap between the two fault segments. Under this configuration, results from our parametric study indicate that successful dynamic triggering of rupture on the Monte Vettore fault is generally inhibited at low pre-stress levels ($S > 0.1$) (Figure 4). Only limited, self-arresting ruptures are triggered at high pre-stress levels ($S = 0.1$), for gap distances of approximately 3 to 4 km and an along-strike offset of about 2.5 km. Run-away ruptures may occur on the receiver fault, but only under high pre-stress conditions ($S = 0.1$) and at close separations, with gap distances < 3 km and offsets < 2.5 km (Figure 4). These results provide a plausible explanation for why rupture did not propagate across the entire Laga-Monte Vettore fault system in a single event but instead occurred as a sequence of distinct earthquakes.

CONCLUSIONS

Using 468 three-dimensional dynamic rupture simulations of interacting normal faults, assuming planar, parallel fault segments, a linear slip-weakening friction law, and different regional stress conditions, we investigated the physical conditions controlling rupture jumps across step-overs. The main findings of this study can be summarized as follows:

- The rupture jump potential in normal fault systems is primarily controlled by the pre-stress level, the relative position of fault segments, and the regional stress orientation, with pre-stress appearing to be the dominant factor.

- Successful rupture jumps can occur across step-overs of up to 5 km when overlap exists between fault segments, even at relatively low pre-stress levels ($S = 0.3$).

- In the absence of fault overlap, the maximum critical distance for successful rupture jumps decreases to approximately 3 km under the tested configurations.

- The relative position of the receiver fault strongly influences rupture behavior. Triggered ruptures are more likely to occur and to propagate successfully and evolve into run-away ruptures when the receiver fault is located on the hanging-wall of the emitter fault than when it is located on the footwall.

- Fault overlap promotes rupture jump viability but may reduce the final earthquake magnitude through stress-shadowing effects that limit the development of the triggered rupture on the receiver fault.

- Dynamic rupture simulations reveal that positive static Coulomb stress changes do not necessarily result in successful rupture jumps. Static Coulomb stress analysis can therefore identify regions favorable for rupture transfer, but does not alone predict whether a rupture jump will occur or whether an initiated rupture will self-arrest or develop into a run-away rupture.

- Under the investigated configurations, the results suggest that approximately 5 km represents an upper bound for rupture jumps in normal fault systems, consistent with limits previously proposed for other faulting styles.

DATA AND RESOURCES

As an effort toward open science and reproducibility, all SeisSol input files, meshes, output data, and scripts required to reproduce the results and figures presented in this manuscript are available at: https://doi.org/10.5281/zenodo.21906008. Supplementary Material accompanies this work, providing a complete list of the resulting scenarios (i.e., jump, no jump, self-arrested, or runaway ruptures) for all 468 configurations considered (Tables S3 to S8). It also includes complementary summary figures (S1 to S5) and two videos illustrating the hanging-wall (HW) and footwall (FW) cases discussed in Section Results. While these additional materials offer further context and visualization, they are not essential

for understanding this research work. The configuration of the simplified model described in Section Model: geometry and physical conditions is selected based on a fault database from the Central Italy region (i.e. Fault2SHA, Faure Walker et al., 2021). This simplified model was developed with reference to the orientations of the Monte Vettore and Laga faults, which were involved in the 2016 Amatrice-Visso-Norcia earthquake sequence (Improta et al., 2019). The alongside aim of this model is to explore the conditions that may have prevented a single large rupture event across more than one fault segment, resulting instead in three significant earthquakes (24/08/16 Mw6.2 Amatrice, 26/10/16 Mw6.1 Visso and 30/10/16 Mw6.6 Norcia earthquakes, USGS catalog last accessed on 01/11/25). All meshes were built using SimModeler software by Simmetrix. We use the open-source software SeisSol v202103 (Gabriel et al., 2025). Some plots were made using Matplotlib v3.7.1 and Python v3.8.13.

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REFERENCES

Ando, R. and Kaneko, Y. (2018). Dynamic rupture simulation reproduces spontaneous multifault rupture and arrest during the 2016 mw 7.9 kaikoura earthquake. *Geophysical Research Letters*, 45(23):12–875.

Bai, K. and Ampuero, J.-P. (2017). Effect of seismogenic depth and background stress on physical limits of earthquake rupture across fault step-overs. *JGR Solid Earth*, 122(12):10–280.

Barall, M. (2009). A grid-doubling finite-element technique for calculating dynamic three-dimensional spontaneous rupture on an earthquake fault. *Geophysical Journal International*, 178(2):845–859.

Bazán Flores, L. I., Kyriakopoulos, C., Oglesby, D. D., Meltzner, A. J., Rockwell, T. K., Fletcher, J. M., and Brothers, D. (2024). Exploring the dynamic interactions between the southern san andreas fault and a normal fault under the salton sea. *Journal of Geophysical Research: Solid Earth*, 129(11):e2023JB028621.

Bell, R., McNeill, L., Bull, J., Henstock, T., Collier, R. L., and Leeder, M. (2009). Fault architecture, basin structure and evolution of the gulf of corinth rift, central greece. *Basin Research*, 21(6):824–855.

Bernard, P. and Zollo, A. (1989). The irpinia (italy) 1980 earthquake: detailed analysis of a complex normal faulting. *JGR Solid Earth*, 94(B2):1631–1647.

Bruhat, L., Fang, Z., and Dunham, E. M. (2016). Rupture complexity and the supershear transition on rough faults. *Journal of Geophysical Research: Solid Earth*, 121(1):210–224.

744 Chiaraluce, L., Di Stefano, R., Tinti, E., Scognamiglio, L., Michele, M., Casarotti, E.,
745 Cattaneo, M., De Gori, P., Chiarabba, C., Monachesi, G., et al. (2017). The 2016 central
746 italy seismic sequence: A first look at the mainshocks, aftershocks, and source models.
747 *Seismological Research Letters*, 88(3):757–771.

748 Oeser, J., Bunge, H.-P., and Mohr, M. (2006). Cluster design in the Earth sciences: Tethys.
749 *Lecture Notes in Computer Science (including subseries Lecture Notes in Artificial Intel-*
750 *ligence and Lecture Notes in Bioinformatics)*.

751 Duman, T. Y., Emre, O., Dogan, A., and Ozalp, S. (2005). Step-over and bend structures
752 along the 1999 duzce earthquake surface rupture, north anatolian fault, turkey. *Bulletin*
753 *of the Seismological Society of America*, 95(4):1250–1262.

754 Ebinger, C. J. and Sleep, N. (1998). Cenozoic magmatism throughout east africa resulting
755 from impact of a single plume. *Nature*, 395(6704):788–791.

756 Faure Walker, J., Boncio, P., Pace, B., Roberts, G., Benedetti, L., Scotti, O., Visini, F.,
757 and Peruzza, L. (2021). Fault2SHA Central Apennines database and structuring active
758 fault data for seismic hazard assessment. *Scientific data*, 8(1):87.

759 Finzi, Y. and Langer, S. (2012). Damage in step-overs may enable large cascading earth-
760 quakes. *Geophysical Research Letters*, 39(16).

761 Freed, A. M. (2005). Earthquake triggering by static, dynamic, and postseismic stress
762 transfer. *Annu. Rev. Earth Planet. Sci.*, 33(1):335–367.

763 Gabriel, A.-A., Kurapati, V., Niu, Z., Schliwa, N., Schneller, D., Ulrich, T., et al. (2025).
764 SeisSol (v1.3.2). Zenodo.

765 Galvez, P., Somerville, P., Petukhin, A., Ampuero, J.-P., and Peter, D. (2020). Earthquake
766 cycle modelling of multi-segmented faults: dynamic rupture and ground motion simula-

767 tion of the 1992 m w 7.3 landers earthquake. *Pure and Applied Geophysics*, 177(5):2163–
768 2179.

769 Ghisetti, F. and Vezzani, L. (2002). Normal faulting, extension and uplift in the outer
770 thrust belt of the central apennines (italy): role of the caramanico fault. *Basin Research*,
771 14(2):225–236.

772 Gupta, A. and Scholz, C. H. (2000). A model of normal fault interaction based on obser-
773 vations and theory. *Journal of structural Geology*, 22(7):865–879.

774 Hamling, I. J., Hreinsdóttir, S., Clark, K., Elliott, J., Liang, C., Fielding, E., Litchfield, N.,
775 Villamor, P., Wallace, L., Wright, T. J., et al. (2017). Complex multifault rupture during
776 the 2016 m w 7.8 kaikōura earthquake, new zealand. *Science*, 356(6334):eaam7194.

777 Harris, R. A., Barall, M., Aagaard, B., Ma, S., Roten, D., Olsen, K., Duan, B., Liu, D.,
778 Luo, B., Bai, K., et al. (2018). A suite of exercises for verifying dynamic earthquake
779 rupture codes. *Seismological Research Letters*, 89(3):1146–1162.

780 Hikima, K. (2012). 2011 4 11 (mj7. 0) — 2 —. *2* , 64(4):243–256.

781 Hollingsworth, J., Ye, L., and Avouac, J.-P. (2017). Dynamically triggered slip on a splay
782 fault in the mw 7.8, 2016 kaikoura (new zealand) earthquake. *Geophysical Research*
783 *Letters*, 44(8):3517–3525.

784 Hu, F., Oglesby, D. D., Zhang, W., and Lu, Z. (2025). Bi-material effects on crit-
785 ical jump distance over step-overs. *Journal of Geophysical Research: Solid Earth*,
786 130(6):e2024JB030992.

787 Idini, B. and Ampuero, J.-P. (2022). Rupture complexity promoted by damaged fault zones
788 in earthquake cycle models. *Authorea Preprints*.

789 Improta, L., Latorre, D., Margheriti, L., Nardi, A., Marchetti, A., Lombardi, A. M.,
790 Castello, B., Villani, F., Ciaccio, M. G., Mele, F. M., et al. (2019a). Multi-segment

rupture of the 2016 amatrice-visso-norcia seismic sequence (central italy) constrained by the first high-quality catalog of early aftershocks. *Sci. Rep.*, 9(1):1–13.

Improta, L., Latorre, D., Margheriti, L., Nardi, A., Marchetti, A., Lombardi, A. M., Castello, B., Villani, F., Ciaccio, M. G., Mele, F. M., et al. (2019). Multi-segment rupture of the 2016 amatrice-visso-norcia seismic sequence (central italy) constrained by the first high-quality catalog of early aftershocks. *Scientific Reports*, 9(1):6921.

King, G. C., Stein, R. S., and Lin, J. (1994). Static stress changes and the triggering of earthquakes. *Bulletin of the Seismological Society of America*, 84(3):935–953.

Kroll, K. A., Dieterich, J. H., Richards-Dinger, K. B., and Oglesby, D. D. (2023). 3-d simulations of earthquakes rupture jumps: 1. homogeneous pre-stress conditions. *Geophysical Journal International*, 234(1):395–403.

Marschall, E. and Douilly, R. (2025). Effects of dip angle on rupture propagation along branch fault systems. *Bulletin of the Seismological Society of America*, 115(1):54–68.

Nicol, A., Begg, J., Saltogianni, V., Mouslopoulou, V., Oncken, O., and Howell, A. (2023). Uplift and fault slip during the 2016 kaikōura earthquake and late quaternary, kaikōura peninsula, new zealand. *New Zealand Journal of Geology and Geophysics*, 66(2):263–278.

Luo, Y., Ampuero, J. P., Galvez, P., Van Den Ende, M., and Idini, B. (2017). Qdyn: a quasi-dynamic earthquake simulator (v1. 1). *Zenodo*.

Meng, L., Ampuero, J.-P., Stock, J., Duputel, Z., Luo, Y., and Tsai, V. (2012). Earthquake in a maze: Compressional rupture branching during the 2012 m w 8.6 sumatra earthquake. *science*, 337(6095):724–726.

Milner, K. R., Shaw, B. E., and Field, E. H. (2022). Enumerating plausible multifault ruptures in complex fault systems with physical constraints. *Bulletin of the Seismological Society of America*, 112(4):1806–1824.

Oskin, M. E., Arrowsmith, J. R., Corona, A. H., Elliott, A. J., Fletcher, J. M., Fielding, E. J., Gold, P. O., Garcia, J. J. G., Hudnut, K. W., Liu-Zeng, J., et al. (2012). Near-field deformation from the el mayor–cucapah earthquake revealed by differential lidar. *Science*, 335(6069):702–705.

Peshette, P., Lozos, J., Yule, D., and Evans, E. (2019). Dynamic rupture modeling to investigate the role of fault geometry in jumping rupture between parallel-trace thrust faults. *Bulletin of the Seismological Society of America*, 109(6):2168–2186.

Poulimenos, G. (2000). Scaling properties of normal fault populations in the western corinth graben, greece: implications for fault growth in large strain settings. *Journal of Structural Geology*, 22(3):307–322.

Rice, J. R. (1992). Fault stress states, pore pressure distributions, and the weakness of the san andreas fault. In *International geophysics*, volume 51, pages 475–503. Elsevier.

Rodriguez Piceda, C., Mildon, Z. K., van den Ende, M., Ampuero, J.-P., and Andrews, B. J. (2025). Normal fault interactions in seismic cycles and the impact of fault network geometry. *Journal of Geophysical Research: Solid Earth*, 130(4):e2024JB030382.

Shi, Z. and Day, S. M. (2013). Rupture dynamics and ground motion from 3-d rough-fault simulations. *Journal of Geophysical Research: Solid Earth*, 118(3):1122–1141.

Stein, R. S. (1999). The role of stress transfer in earthquake occurrence. *Nature*, 402(6762):605–609.

Tago, J., Cruz-Atienza, V. M., Virieux, J., Etienne, V., and Sánchez-Sesma, F. J. (2012). A 3d hp-adaptive discontinuous galerkin method for modeling earthquake dynamics. *Journal of Geophysical Research: Solid Earth*, 117(B9).

Tanaka, M., Asano, K., Iwata, T., and Kubo, H. (2014). Source rupture process of the 2011

fukushima-ken hamadori earthquake: how did the two subparallel faults rupture? *Earth, Planets and Space*, 66:1–8.

Taufiqurrahman, T., Gabriel, A.-A., Ulrich, T., Valentova, L., and Gallovič, F. (2022). Broadband dynamic rupture modeling with fractal fault roughness, frictional heterogeneity, viscoelasticity and topography: The 2016 mw 6.2 amatrice, italy earthquake. *Geophysical Research Letters*, 49(22):e2022GL098872.

Tinti, E., Casarotti, E., Ulrich, T., Taufiqurrahman, T., Li, D., and Gabriel, A.-A. (2021). Constraining families of dynamic models using geological, geodetic and strong ground motion data: The mw 6.5, october 30th, 2016, norcia earthquake, italy. *Earth and Planetary Science Letters*, 576:117237.

Ulrich, T., Gabriel, A.-A., Ampuero, J.-P., and Xu, W. (2019a). Dynamic viability of the 2016 mw 7.8 kaikōura earthquake cascade on weak crustal faults. *Nat. Commun.*, 10(1):1–16.

Ulrich, T., Vater, S., Madden, E. H., Behrens, J., van Dinther, Y., Van Zelst, I., Fielding, E. J., Liang, C., and Gabriel, A.-A. (2019b). Coupled, physics-based modeling reveals earthquake displacements are critical to the 2018 palu, sulawesi tsunami. *Pure App. Geophys.*, 176(10):4069–4109.

Uphoff, C., Rettenberger, S., Bader, M., Madden, E. H., Ulrich, T., Wollherr, S., and Gabriel, A.-A. (2017). Extreme scale multi-physics simulations of the tsunamigenic 2004 sumatra megathrust earthquake. In *Proc. Int. HPC*, pages 1–16.

Valoroso, L., Chiaraluce, L., Piccinini, D., Di Stefano, R., Schaff, D., and Waldhauser, F. (2013). Radiography of a normal fault system by 64,000 high-precision earthquake locations: The 2009 l'aquila (central italy) case study. *JGR Solid Earth*, 118(3):1156–1176.

Wei, S., Fielding, E., Leprince, S., Sladen, A., Avouac, J.-P., Helmberger, D., Hauksson, E., Chu, R., Simons, M., Hudnut, K., et al. (2011). Superficial simplicity of the 2010 el mayor–cucapah earthquake of baja california in mexico. *Nature geoscience*, 4(9):615–618.

Wernicke, B., Axen, G. J., and Snow, J. K. (1988). Basin and range extensional tectonics at the latitude of las vegas, nevada. *Geological Society of America Bulletin*, 100(11):1738–1757.

Wesnousky, S. G. (2006). Predicting the endpoints of earthquake ruptures. *Nature*, 444(7120):358–360.

Zielke, O. and Mai, P. M. (2023). Mcqsim: A multicycle earthquake simulator. *Bulletin of the Seismological Society of America*, 113(3):889–908.

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Alt-text: 3D schematic view of the digitized fault model used in the simulations. The figure shows the emitter and receiver faults embedded in a layered medium, with their relative geometry defined by along-strike offset (±5 km) and across-strike gap (±5 km). The receiver fault is located to the north of the emitter fault. Both faults have a surface of 10 km × 10 km. Arrows indicate different orientations of the maximum horizontal regional stress (SHmax = 330°, 340°, and 350°). 46

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941 strike. Circles indicate the rupture outcome for each scenario: filled circles
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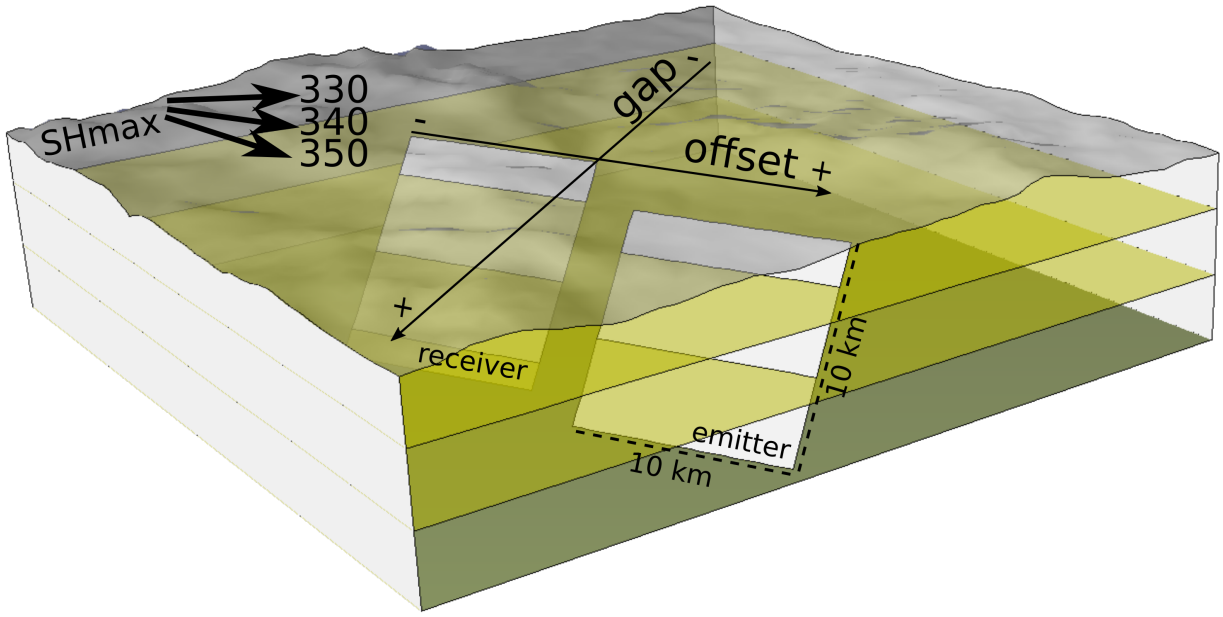


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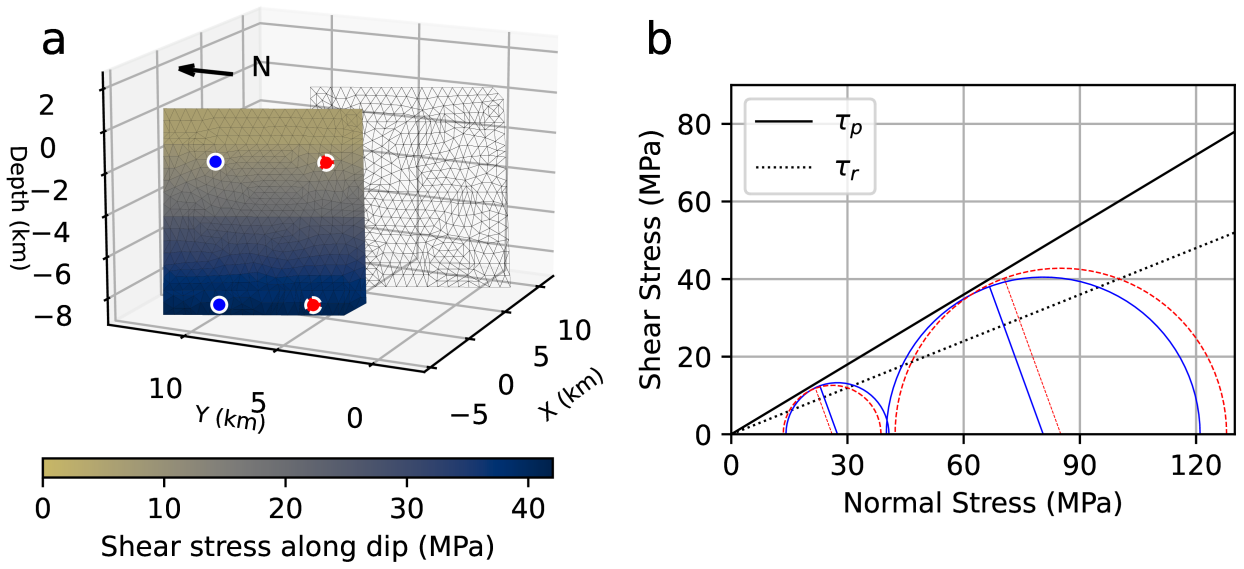


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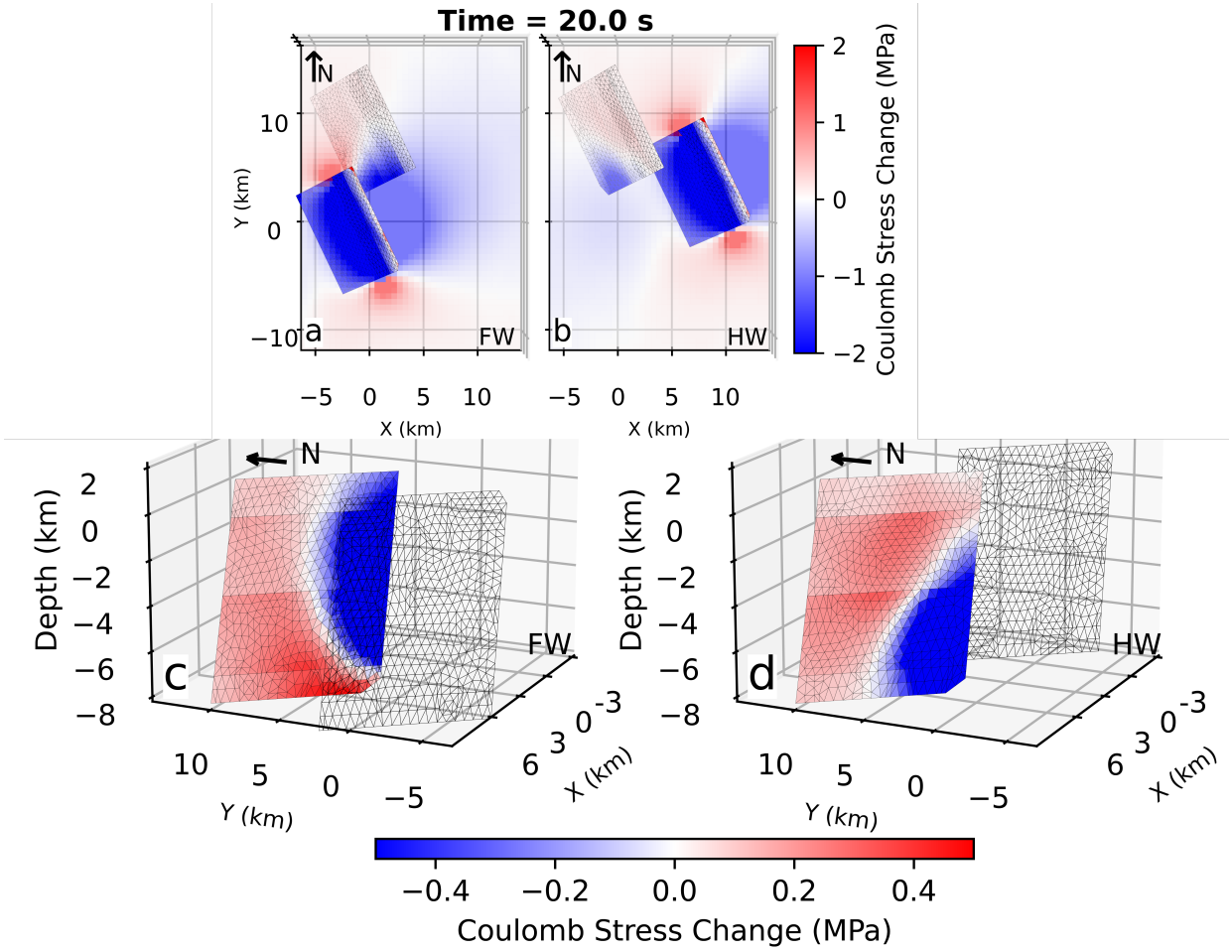


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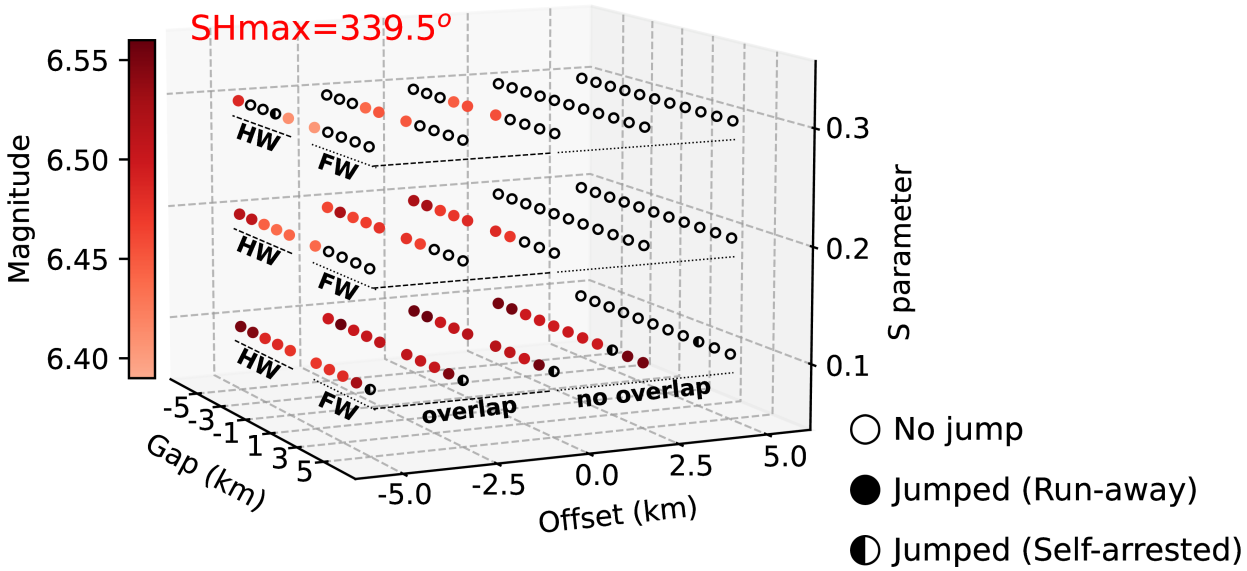


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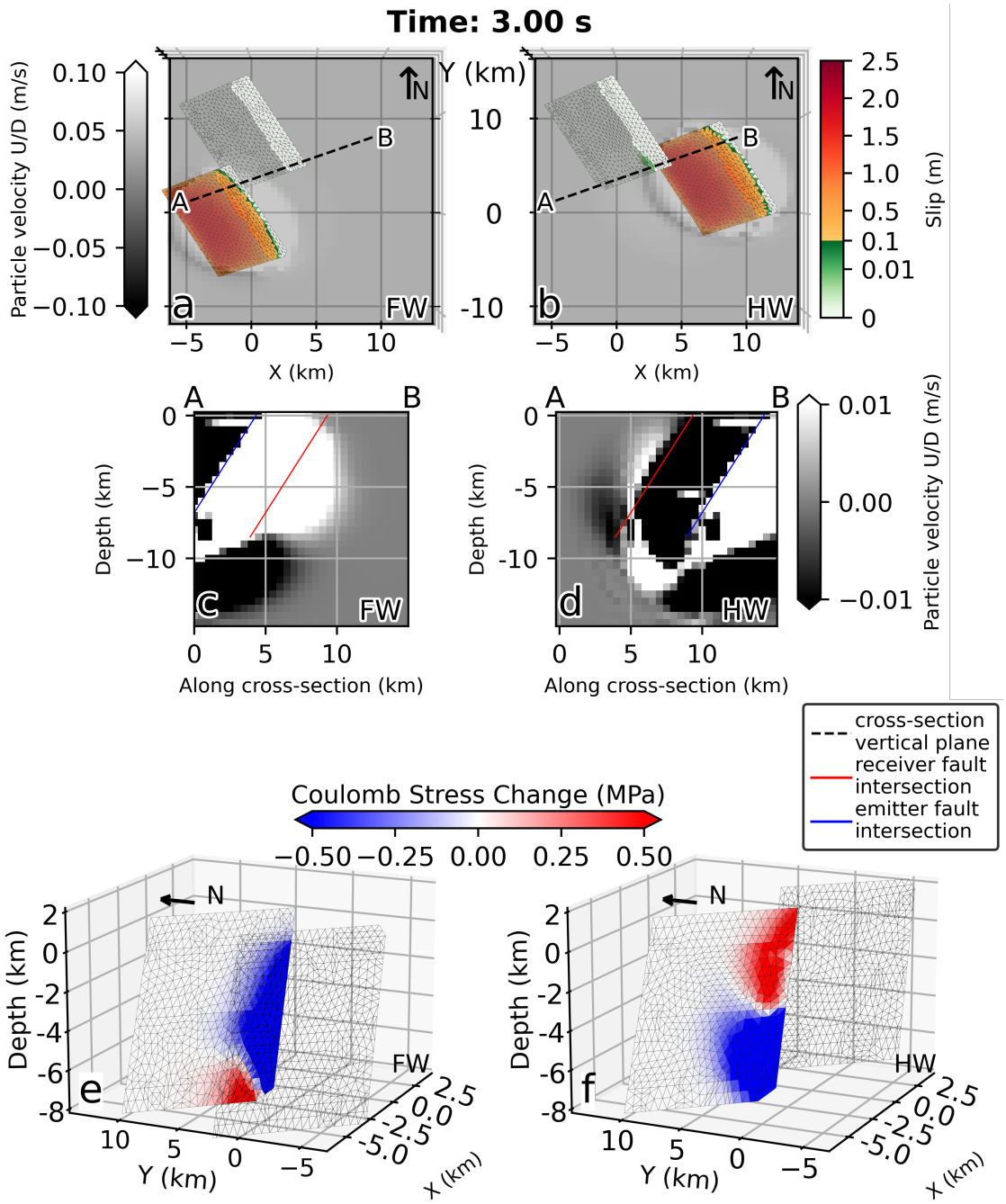


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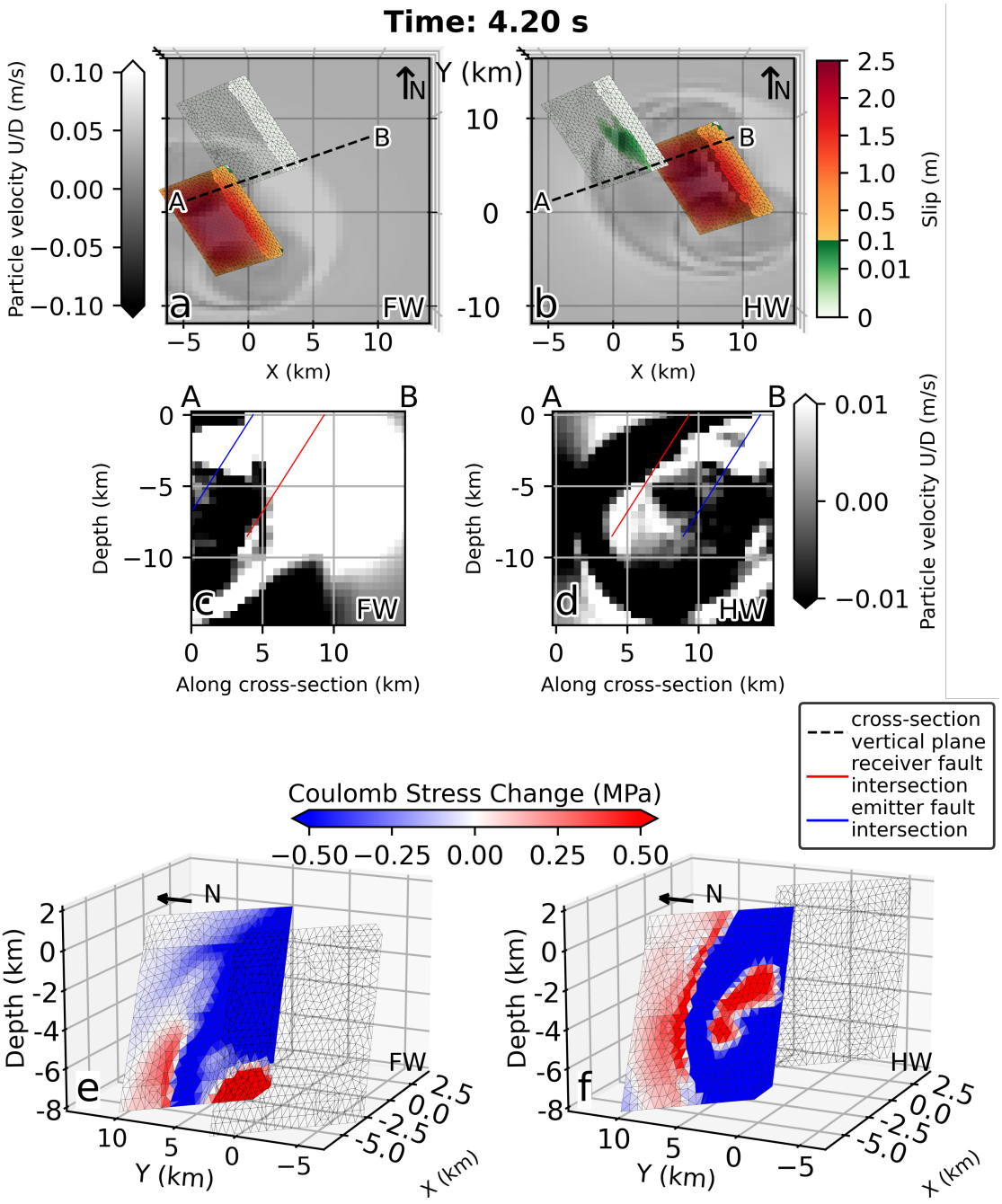


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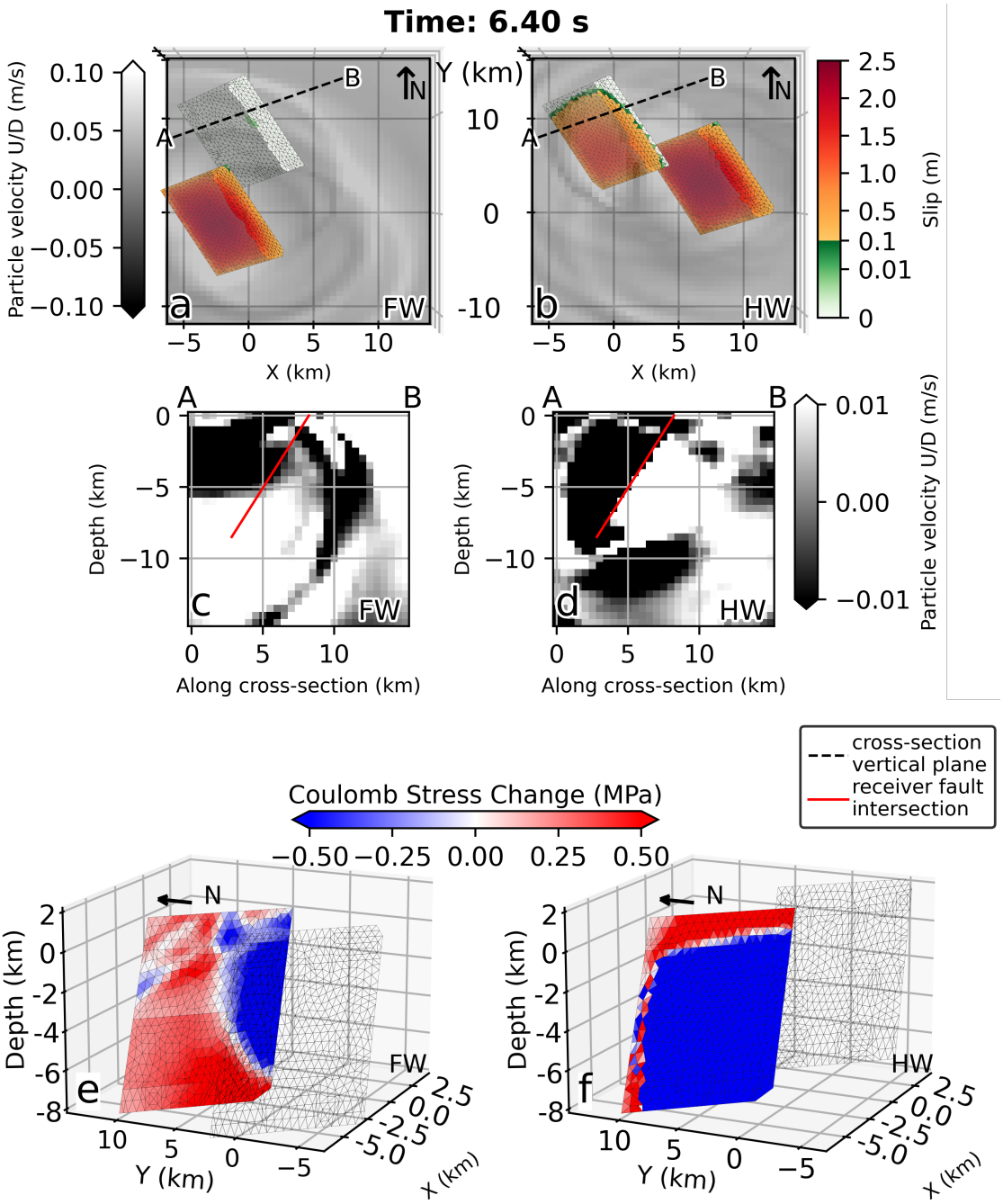


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Parameter	lower bound	increment	upper bound
Gap distance (km)	-5.0	1.0	5.0
Offset distance (km)	-5.0	2.5	5.0
SH_{max} (^{o})	334.5	5	344.5
S	0.1	0.1	0.3
Parameter	Fixed value	Parameter	Fixed value
μ_s	0.6	Fault size (km^2)	10×10
μ_d	0.4	d_c (m)	0.15